Soumil Iyer

(765) 694-9635 | iyer144@purdue.edu | linkedin.com/in/soumil-iyer | soumiliyer.engineer | Available Jan 2027

EDUCATION

Purdue University, West Lafayette, IN

Master of Science in Mechanical Engineering (GPA: 4.00) — January 2026 - December 2026 (Expected)

Bachelor of Science in Mechanical Engineering (GPA: 3.38) — August 2022 - December 2025

EXPERIENCE

Rapid Prototyping Lab Student Employee | Purdue University — **August 2026 - Present**

- Guide CAD-to-part FDM/FFF workflows in Bambu Studio/Cura; select printers, materials, orientation, supports, infill, and layer settings
- Set up and calibrate printers; perform preventive maintenance and troubleshoot process/equipment issues to sustain reliability and consistency
- Review CAD/slicer setups and diagnose geometry, material, and equipment issues; guide changes to improve print quality and limit rework

Graduate Teaching Assistant - Intro to Manufacturing | Purdue University — **January 2026 - May 2026**

- Instructed 19 students in CNC/manual turning and milling, metalworking, electronics, assembly, and DFM from CAD to finished parts
- Taught Y14.5 GD&T/tolerancing and metrology; reviewed drawings, BOMs, and assembly procedures against specs for manufacturability and fit
- Supervised machine use, inspections, safety, and design validation while troubleshooting fabrication, dimensional, and assembly issues

M.E. Undergraduate Research Intern | Purdue University — **June 2025 - August 2025**

- Manufactured 60+ dogbones and 24 disks across 6 thermoplastics in horizontal/vertical orientations for controlled material/process comparison
- Performed ASTM D638 tensile tests and ultrasonic NDE to quantify anisotropy and destructive versus nondestructive material response
- Analyzed stress-strain and ultrasonic wave-speed data in MATLAB to calculate Young's modulus and compare build-orientation effects
- Documented fabrication, test methods, uncertainty, and results showing build orientation significantly affected tensile modulus and failure behavior

M.E. Undergraduate Research Assistant - Energy Storage Analytics | Purdue University — **August 2024 - December 2024**

- Built a Python/customtkinter tool with file upload, analysis, and plotting workflows to process and visualize 1,000+ Li-ion battery datapoints
- Developed reusable import, comparison, and plotting routines to standardize recurring battery-data analysis and research documentation
- Interpreted battery-performance trends to build practical knowledge of Li-ion cell behavior, design considerations, and optimization

Teaching Assistant - Thermodynamics I & ENGR 131 | Purdue University — **August 2024 - December 2024**

PROJECTS

Design of Experiments - Lead-Free Solder Joint Process Optimization | IE 533 — **March 2026 - April 2026**

- Designed randomized 2^3 Minitab DOE for SAC305 shear strength using 230/260 C, 0.5/1.0 mg paste, and 1/4 C/s cooling; 16 simulated runs
- Planned 2 replicates per treatment with randomized run order and controlled nuisance factors; obtained ~0.83 power at alpha=0.05
- Fit GLM/ANOVA (p=0.006) and checked Pareto, interaction, and residual plots; identified 3 significant main effects in simulated data
- Quantified effects: temp +6.90, cooling +5.46, paste +3.36 N/mm^2; recommended 260 C, 1.0 mg, 4 C/s while documenting simulated-data limits

Mechanical Engineering Senior Design - Assisted Eating Utensil — **August 2025 - December 2025**

- Led CAD modeling and FDM prototyping within a team developing a passive, bearing-based self-leveling utensil for users with hand tremors
- Translated user feedback into a 150 mm x 35 mm ribbed handle and finger/wrist straps to improve grip and reduce accidental drops
- Iterated bearing/utensil interfaces and detachable stainless-steel head geometry, balancing self-leveling, scooping, cleaning, and assembly
- Supported team validation: survived 2 m drops without cracks and 14 one-hour dishwasher cycles; assessed head insertion/removal forces
- Contributed to a team prototype costing ~$13 in components, below the $50 cost target and supporting a sub-$25 selling-price goal

Autonomous Robot Control & Line Following | ME 375 — **April 2025 - May 2025**

- Developed the motor speed-difference controller using MATLAB root locus; supported integration of Simulink logic and robot hardware
- Collaborated on rest, race, recovery, and parking logic with PID/PI, encoder and line/IR feedback, and static-friction compensation
- Supported motor/sensor characterization (IR error <0.5 cm over 10-80 cm) and testing for two line-following laps and 30 cm wall parking

Heat Transfer Analysis - Blanket Insulation Materials | ME 315 — **March 2025 - April 2025**

- Instrumented polyester/wool/down with thermocouples/LabVIEW; logged 1-s data for 1-hour cooling runs from 37 C toward ~5 C
- Verified lumped-capacitance validity at Bi ~0.03; linearized transients and calculated k=0.390/0.0152/0.0344 W/m-K for polyester/wool/down

Wind Tunnel Aerodynamic Analysis - Race Car Spoiler | ME 30801 — **March 2025 - April 2025**

- Designed/FDM-printed a SolidWorks spoiler; calibrated LVDT/load-cell setup and tested 7 angles (0-60 deg) at 30.880 m/s, 3 trials each
- Analyzed 15-s force datasets in MATLAB with standard-deviation error bars; found peak Cl ~0.65 at 40 deg and evaluated drag tradeoffs

Autonomous Rover Development - Drone Video VIP | VIP 27920 — **August 2023 - December 2023**

- Supported rover hardware/software integration for miniature-city navigation using visual detection, IR, and LiDAR in a drone-filming system
- Trained/evaluated a CIFAR-10 model in Python/Google Colab; achieved 78.74% accuracy after 10 epochs and planned ResNet comparisons

SKILLS & CERTIFICATIONS

CAD & Design: CATIA V5/V6, Siemens NX (PLM/PDM), SolidWorks, Fusion 360, AutoCAD, Mastercam, tolerance stack-up
Mechanical Analysis: Solid mechanics, mechanics of materials, structural/thermal FEA, heat transfer (ME 323/32301/315/354)
Design & Reliability: GD&T (ANSI/ASME Y14.5), DFM/DFA, engineering drawings, BOMs, DFMEA, PFMEA, Fault Tree Analysis (FTA)
Materials: Engineering metals/polymers, structure-property relationships, material selection, ASTM D638 characterization (MSE 230; research)
Manufacturing Processes: CNC/manual machining, FDM/FFF; stamping, deep drawing, injection molding, heat treating, welding fundamentals
Testing & Data: DoE, ANOVA/GLM, Minitab, MATLAB, LabVIEW, DAQ, metrology, thermocouples, LVDTs, NDE, uncertainty analysis
Programming & Controls: Python, MATLAB, C, Simulink, ROBOGUIDE, Studio 5000, Ladder Logic, PID/PI, root locus, sensor calibration
Lean & Manufacturing Systems: Lean, Kaizen, 6S, Pareto analysis, 5 Why, Fishbone diagrams
Certifications: SolidWorks Milestones (in progress); SOLIDWORKS certification (exam pending); FEA; Simulink; MATLAB; Python; Arduino